Report Submission Date:

Name:		**Date:**	
MRN:		**Modality Type:**	MR
Gender:		**Description:**	MR CERVICAL SPINE WO
DOB:		**Laterality:**	
		Referrer:	Second Opinions

MRI cervical spine without contrast.

CLINICAL HISTORY: Reason for Initial Exam Pain, heaviness in arms and hands.

TECHNIQUE: Fast spin echo T2 and spin echo T1 sequences were obtained in axial and sagittal planes.

Findings:

Previous MRIs dated 04/10/23 and 1/15/21 is available for comparison.

There is straightening of cervical lordosis.

There is mild chronic anterior wedging of C5 vertebral body.

The rest of the osseous elements are intact with no evidence of fracture or dislocation.

The cervical cord is of normal uniform signal intensity without evidence of focal expansion.

There is no evidence for tonsillar herniation. Limited views of the posterior fossa reveal no abnormalities.

Multilevel disk dehydration, reduced disc heights and marginal osteophytes are seen.

At C2-C3, there is a 10 x 1.5 mm disc herniation showing an annular tear indenting the ventral thecal sac causing minimal left neural foramina narrowing. Canal is patent.

At C3-C4, there is a 11 x 3 mm disc herniation indenting the ventral thecal sac with uncovertebral hypertrophy causing moderate left and mild-to-moderate right neural foramina narrowing. Canal is borderline stenotic.

At C4-C5, there is a 12 x 3 mm disc herniation indenting the ventral thecal sac with uncovertebral hypertrophy causing mild right and mild-to-moderate left neural foramina narrowing. Canal is mildly stenotic.

At C5-C6, there is a 11 x 2.5 mm disc herniation indenting the ventral thecal sac with uncovertebral hypertrophy causing mild right and minimal left neural foramina narrowing. Canal is borderline stenotic.

At C6-C7, there is a 12 x 3 mm disc herniation indenting the ventral thecal sac with uncovertebral hypertrophy causing mild-to-moderate right and mild left neural foramina narrowing. Canal is borderline stenotic.

Remaining levels are unremarkable.

IMPRESSION:

1. Straightening of cervical lordosis.

2. At C2-C3, there is a 10 x 1.5 mm disc herniation showing an annular tear indenting the ventral thecal sac causing minimal left neural foramina narrowing. Canal is patent.

3.At C3-C4, there is a 11 x 3 mm disc herniation indenting the ventral thecal sac with uncovertebral hypertrophy causing moderate left and mild-to-moderate right neural foramina narrowing. Canal is borderline stenotic.

4.At C4-C5, there is a 12 x 3 mm disc herniation indenting the ventral thecal sac with uncovertebral hypertrophy causing mild right and mild-to-moderate left neural foramina narrowing. Canal is mildly stenotic.

5.At C5-C6, there is a 11 x 2.5 mm disc herniation indenting the ventral thecal sac with uncovertebral hypertrophy causing mild right and minimal left neural foramina narrowing. Canal is borderline stenotic.

6.At C6-C7, there is a 12 x 3 mm disc herniation indenting the ventral thecal sac with uncovertebral hypertrophy causing mild-to-moderate right and mild left neural foramina narrowing. Canal is borderline stenotic.

7.As compared to the previous MRIs dated 04/10/23 and 1/15/21, there is evidence of small additional disc herniation at C2-C3 level with mild increase in size of the disc herniations at C3-C4 and C4-C5 levels.

SECOND OPINIONS DISCLAIMER:
Note: The above report was provided by the signing radiologist at the request of the referring physician, patient or patient representative as a second opinion. The opinions and recommendations included therein are provided on a strict advisory basis and are based on the images and clinical information provided. Neither the radiologist nor the company assume responsibility for any decisions made or actions taken based on this report nor for any effects that result from such decisions and actions. The responsibility for clinical decisions remains entirely upon the patient's medical team. This information has been clearly communicated at the time of request of this second opinion.

Electronically signed on by:

Fellowship Trained MRI and CT Specialist

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INTEGRITY
DIAGNOSTICS

Test Date: 1/2/2020

Patient:	Sample Radiculopathy Case	**DOB:**	7/7/1947	**Physician:**	Dr. MD
Sex:	Male				

Diagnosis: Cervical radiculopathy M54.12

Complaint:
The patient complains of lower cervical pain and paresthesia in the left shoulder and lateral arm. There are no complaints of right sided symptoms or lower extremity symptoms and past medical history is negative for DM, vitamin deficiency, excessive ETOH, thyroid or other autoimmune disorders. A brief clarifying exam is performed prior to testing and temperature is maintained above 31C in the left upper extremity.

ELECTRONEUROMYOGRAPHIC SUMMARY

Nerve Conduction:

Sensory NCS: The left median, radial and ulnar nerves are WNL.

Motor NCS: The left median, radial and ulnar nerves are WNL.

Late Responses: The left median and ulnar F-waves are WNL.

Electromyography:
Using a monopolar needle electrode, sampling consisted of the dorsal primary rami innervated muscles from the left C5-T1 paraspinals as well as ventral primary rami innervated muscles of the C5-T1 myotomes in the upper extremity. Muscle membrane instability (fibrillation/positive sharp wave activity) was observed in the left C6-7 paraspinals, the left triceps and left FCR muscles. Paraspinal assessment was technically challenging due to decreased patient ability to consistently maintain electrical silence during assessment. All other muscles tested showed normal activity at insertion, rest and during voluntary recruitment.

Impression: (Abnormal Study)

1. **Evidence suggests a mild acute left lower cervical radiculopathy (C6-7) that affects the dorsal and ventral primary rami.**

2. **There is no evidence to suggest a polyneuropathy, plexopathy or focal entrapment.**

INTEGRITY
DIAGNOSTICS
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Nerve Conduction Studies

Anti Sensory Summary Table

Site	NR	Peak (ms)	Norm Peak (ms)	O-P* Amp (µV)	Norm O-P Amp	Site1	Site2	Delta-P (ms)	Dist (cm)	Vel (m/s)	Norm Vel (m/s)
Left Median Anti Sensory (3rd Digit)											
Palm		1.8	<2.2	39.1	>10	Palm	3rd Digit	1.8	7.0	39	
Wrist		3.1	<3.6	32.2		Wrist	Palm	1.3	7.0	54	>39
Left Ulnar Anti Sensory (5th Digit)											
Wrist		2.6	<3.4	22.6	>10	Wrist	5th Digit	2.6	12.0	46	

Motor Summary Table

Site	NR	Onset (ms)	Norm Onset (ms)	O-P* Amp (mV)	Norm O-P Amp	Neg Dur (ms)	Site1	Site2	Delta-0 (ms)	Dist (cm)	Vel (m/s)	Norm Vel (m/s)
Left Median Motor (Abd Poll Brev)												
Wrist 8cm		3.6	<4.2	4.4	>2.8	5.63	Elbow	Wrist 8cm	3.7	21.5	58	>50
Palm		1.9		4.7		5.78	Wrist 8cm	Palm	1.7	5.5	32	>29
Elbow		7.3		4.0		5.94						
Left Radial Motor (Ext Ind Prop)												
Forearm 8cm		2.7	<3.0	7.2		7.66	Brachium	Forearm 8cm	3.2	20.0	63	>50
Brachium		5.9		7.1		7.50						
Left Ulnar Motor (Abd Dig Minimi)												
Wrist 8cm		2.8	<3.6	7.3	>4.2	5.00	B Elbow	Wrist 8cm	2.8	17.0	61	>49
B Elbow		5.6		7.0		5.31	A Elbow	B Elbow	1.9	10.0	53	>47
A Elbow		7.5		6.9		5.47						

Comparison Summary Table

Site	NR	Peak (ms)	O-P* Amp (µV)	Norm O-P Amp	Site1	Site2	Delta-P (ms)
Left Median/Radial Comparison (Digit 1)							
Median 10cm		2.6	23.0	>7uV	Median 10cm	Radial 10 cm	0.1
Radial 10 cm		2.5	7.6	>5uV			

F Wave Studies

NR	F-Lat (ms)	Lat Norm (ms)	L-R F-Lat (ms)	L-R Lat Norm
Left Median (Mrkrs) (Abd Poll Brev)				
	21.71	<33		<2.0
Left Ulnar (Mrkrs) (Abd Dig Min)				
	24.42	<36		<2.0

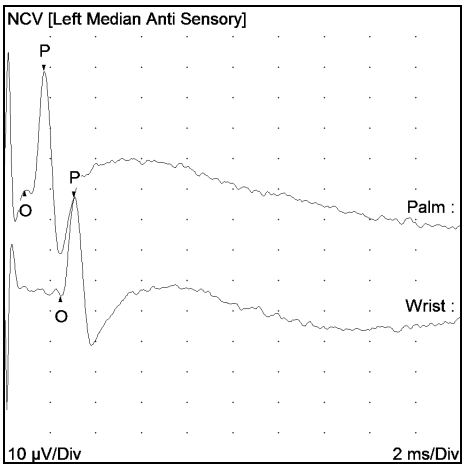
EMG

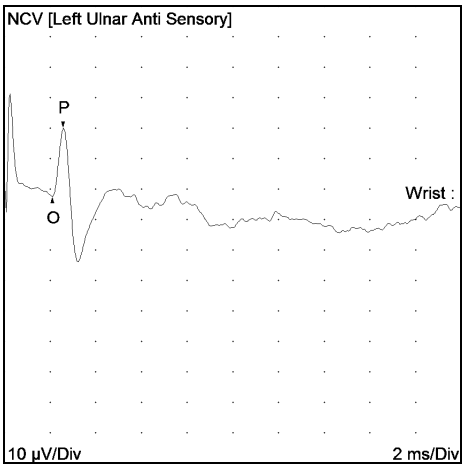
Side	Muscle	Nerve	Root	Ins Act	Fibs	Psw	Other	Amp	Dur	Poly	Recrt	Int Pat	Comment
Left	1stDorInt	Ulnar	C8-T1	Nml	Nml	Nml	Nml	Nml	Nml	0	Nml	Nml	
Left	FlexCarRad	Median	C6-7	**Incr**	**1+**	Nml	Nml	Nml	Nml	0	Nml	Nml	
Left	BrachioRad	Radial	C5-6	Nml	Nml	Nml	Nml	Nml	Nml	0	Nml	Nml	
Left	Triceps	Radial	C6-7-8	**Incr**	Nml	**1+**	Nml	Nml	Nml	0	Nml	Nml	
Left	Deltoid	Axillary	C5-6	Nml	Nml	Nml	Nml	Nml	Nml	0	Nml	Nml	
Left	ExtDigCom	Radial (Post Int)	C7-8	Nml	Nml	Nml	Nml	Nml	Nml	0	Nml	Nml	

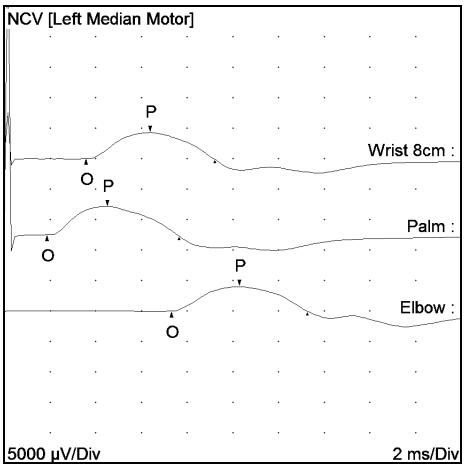
Paraspinal EMG

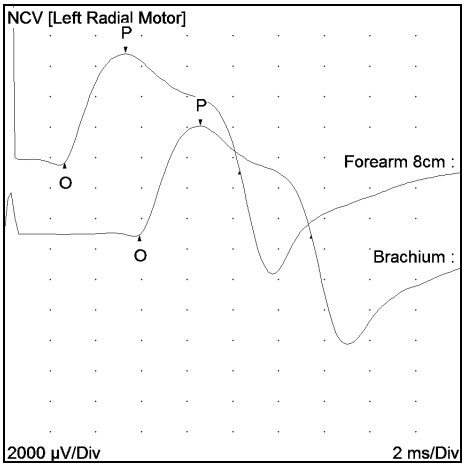
Side	Muscle	Nerve	Root	Ins Act	Fibs	Psw	Other	Amp	Dur	Poly	Recrt	Int Pat	Comment
Left	C5-6	DPR	C5-6	Nml	Nml	Nml	Nml	NT'd	NT'd	NT'd	NT'd	NT'd	
Left	C6-7	DPR	C6-7	**Incr**	**1+**	**1+**	Nml	NT'd	NT'd	NT'd	NT'd	NT'd	
Left	C7-T1	DPR	C7,C8	Nml	Nml	Nml	Nml	NT'd	NT'd	NT'd	NT'd	NT'd	

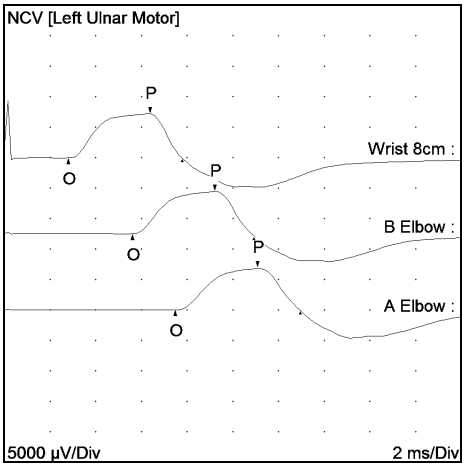
Waveforms:

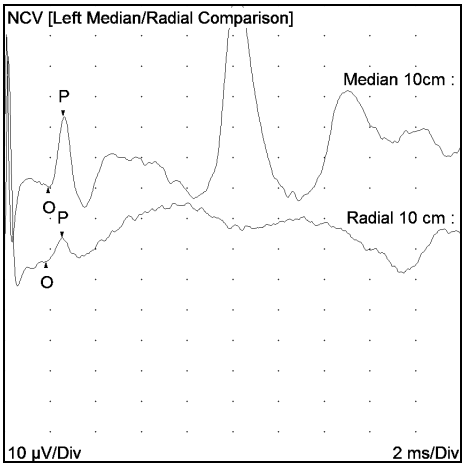

NCV [Left Median Anti Sensory]
P
O
P
Palm :
O
Wrist :
10 µV/Div
2 ms/Div

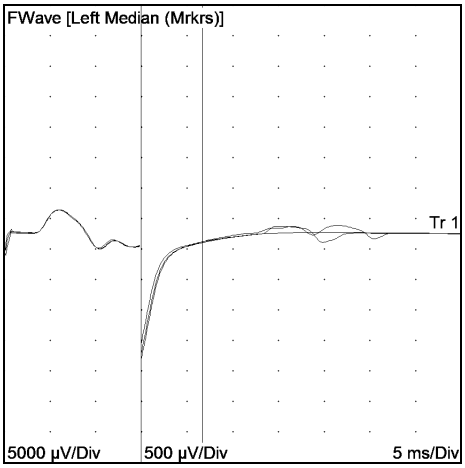

NCV [Left Ulnar Anti Sensory]
P
O
Wrist :
10 µV/Div
2 ms/Div

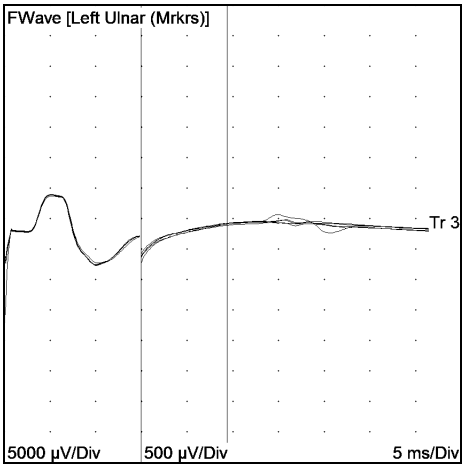

NCV [Left Median Motor]
P
Wrist 8cm :
O
P
Palm :
O
P
Elbow :
O
5000 µV/Div
2 ms/Div

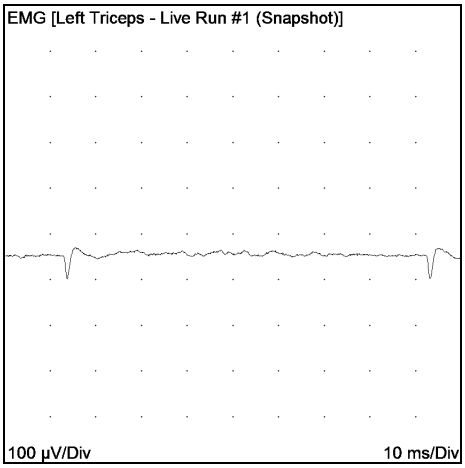

NCV [Left Radial Motor]
P
P
Forearm 8cm :
O
O
Brachium :
2000 µV/Div
2 ms/Div

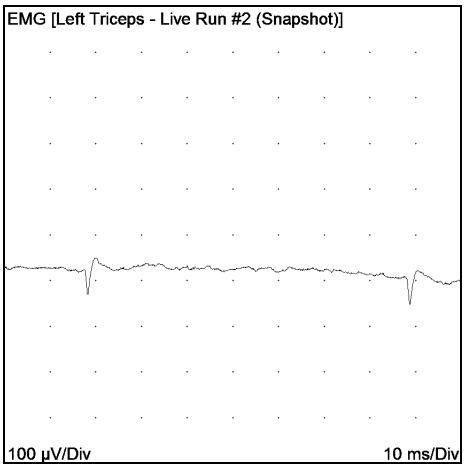

NCV [Left Ulnar Motor]
P
Wrist 8cm :
O
P
B Elbow :
O
P
A Elbow :
O
5000 µV/Div
2 ms/Div

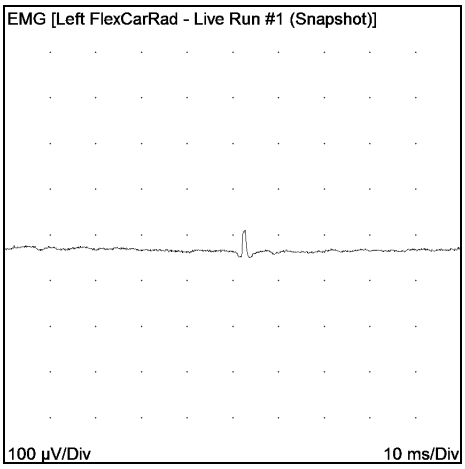

NCV [Left Median/Radial Comparison]
Median 10cm :
P
O
P
Radial 10 cm :
O
10 µV/Div
2 ms/Div

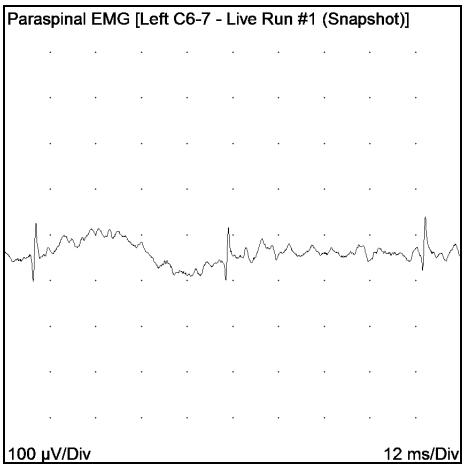

FWave [Left Median (Mrkrs)]
Tr 1
5000 µV/Div
500 µV/Div
5 ms/Div

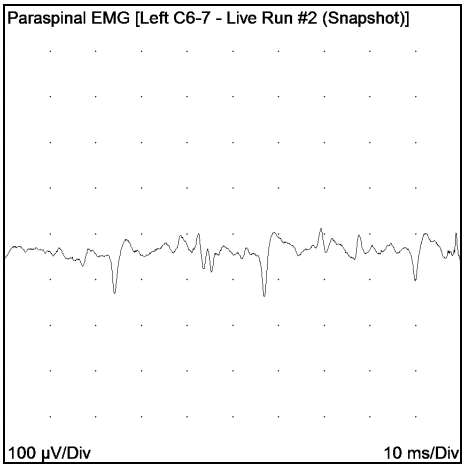

FWave [Left Ulnar (Mrkrs)]
Tr 3
5000 µV/Div
500 µV/Div
5 ms/Div


EMG [Left Triceps - Live Run #1 (Snapshot)]
100 µV/Div
10 ms/Div


EMG [Left Triceps - Live Run #2 (Snapshot)]
100 µV/Div
10 ms/Div


EMG [Left FlexCarRad - Live Run #1 (Snapshot)]
100 µV/Div
10 ms/Div


Paraspinal EMG [Left C6-7 - Live Run #1 (Snapshot)]
100 µV/Div
12 ms/Div


Paraspinal EMG [Left C6-7 - Live Run #2 (Snapshot)]
100 µV/Div
10 ms/Div